

1 (a) Underline **all** the vector quantities in the following list:

distance

displacement

speed

velocity

acceleration.

[1]

- (b) A car of mass 1400 kg is initially travelling horizontally in a straight line at a constant speed of 15 m s^{-1} . The brakes are applied and the car decelerates uniformly to rest in a distance of 35 m.

Calculate:

- (i) the initial kinetic energy of the car

kinetic energy = J [2]

- (ii) the acceleration of the car

acceleration = m s^{-2} [2]

- (iii) the resultant force on the decelerating car.

force = N [2]

- (c) The initial speed of the car in 1(b) is now doubled to 30 m s^{-1} . The brakes of the car are then applied and the car comes to rest. The resultant force on the car remains the same as in 1(b)(iii).

Describe and explain the effect, if any, of doubling the initial speed on:

- (i) the magnitude of the acceleration of the car

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..... [1]

- (ii) the distance travelled by the car in coming to rest.

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..... [2]

[Total: 10]